DSA 210 TERM PROJECT
EDA & HYPOTHESIS TESTS

1. Data Analysis

The analysis in this project was conducted in multiple stages, following a structured data science pipeline including data preprocessing, exploratory data analysis (EDA), statistical testing, and machine learning modeling.

1.1. Data Preprocessing

Initially, the dataset was cleaned and transformed into a suitable format for analysis. Time-related variables such as departure and arrival times were converted into datetime format, and travel durations were standardized in minutes, initial and final data formatsa are given in Figure 2 and Figure 3 respectively.

```
First 5 rows:
                  date departure   arrival actual_duration  \
0  2026-03-31 00:00:00  07:49:00  08:30:00        00:41:00
1  2026-03-31 00:00:00  17:25:00  18:16:00        00:51:00
2  2026-03-31 00:00:00  18:31:00  18:36:00        00:05:00
3  2026-03-31 00:00:00  20:17:00  20:21:00        00:04:00
4  2026-04-01 00:00:00  13:32:00  14:48:00        01:16:00

          destination  num_lights distance estimated_duration trafic_level
0  Florya - Altunizade          13    29 km           00:44:00       Medium
1  Altunizade - Florya           8    31 km           00:53:00         High
2      Florya - Florya           0   0.6 km           00:02:00          Low
3      Florya - Florya           0   1.1 km           00:03:00          Low
4       Florya - Tuzla          11    64 km           01:14:00       Medium
```

Figure 2

Additional variables such as delay (difference between estimated and actual duration) and rush hour indicators were also derived.

```
Final Data:
        date           departure   arrival actual_duration  \
0  2026-03-31 2026-05-05 07:49:00  08:30:00        00:41:00
1  2026-03-31 2026-05-05 17:25:00  18:16:00        00:51:00
2  2026-03-31 2026-05-05 18:31:00  18:36:00        00:05:00
3  2026-03-31 2026-05-05 20:17:00  20:21:00        00:04:00
4  2026-04-01 2026-05-05 13:32:00  14:48:00        01:16:00

          destination  num_lights  distance estimated_duration trafic_level  \
0  Florya - Altunizade          13      29.0           00:44:00       Medium
1  Altunizade - Florya           8      31.0           00:53:00         High
2      Florya - Florya           0       0.6           00:02:00          Low
3      Florya - Florya           0       1.1           00:03:00          Low
4       Florya - Tuzla          11      64.0           01:14:00       Medium

   actual_duration_min  estimated_duration_min rush_hour  delay_min
0                 41.0                    44.0       Yes       -3.0
1                 51.0                    53.0       Yes       -2.0
2                  5.0                     2.0       Yes        3.0
3                  4.0                     3.0        No        1.0
4                 76.0                    74.0        No        2.0
```

Figure 3

The final dataset consisted of 127 observations and 9 variables, with no missing values, ensuring data completeness and reliability for further analysis.

```
Shape: (127, 9)

Columns:
Index(['date', 'departure', 'arrival', 'actual_duration', 'destination',
       'num_lights', 'distance', 'estimated_duration', 'trafic_level'],
      dtype='object')

Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 127 entries, 0 to 126
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   date                127 non-null    object
 1   departure           127 non-null    object
 2   arrival             127 non-null    object
 3   actual_duration     127 non-null    object
 4   destination         127 non-null    object
 5   num_lights          127 non-null    int64
 6   distance            127 non-null    object
 7   estimated_duration  127 non-null    object
 8   trafic_level        127 non-null    object
dtypes: int64(1), object(8)
memory usage: 9.1+ KB
None
```

Figure 4

```
Missing values:
date                  0
departure             0
arrival               0
actual_duration       0
destination           0
num_lights            0
distance              0
estimated_duration    0
trafic_level          0
dtype: int64
```

Figure 5

1.2. Exploratory Data Analysis (EDA)

1.2.1. Distribution Histograms

The distribution of actual travel duration is right-skewed as shown in Figure 6, meaning that most trips are relatively short, while a small number of trips take significantly longer. The majority of observations are concentrated between approximately 10 to 60 minutes, indicating that typical daily trips are of moderate duration. However, a few extreme values (outliers) extend the distribution toward higher durations, suggesting occasional long-distance or high-delay trips.

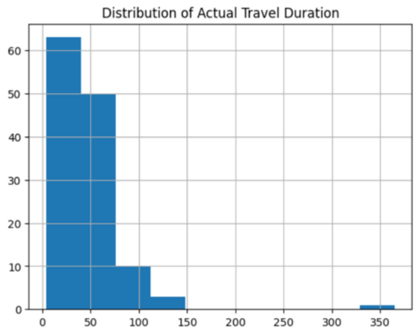

Figure 6

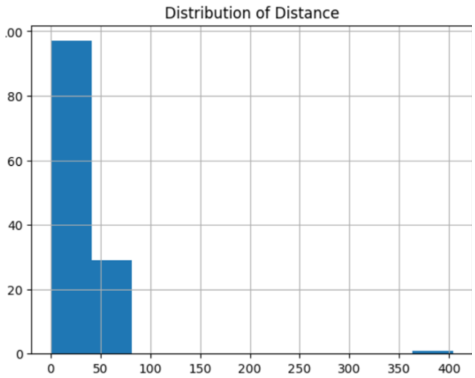

Figure 7

The distance distribution also exhibits a right-skewed pattern, similar to travel duration. Most trips fall within a shorter distance range (roughly below 50 km), while a very small number of trips reach much higher distances. This indicates that the dataset is dominated by short to medium-range trips, with only a few long-distance journeys.

This pattern is consistent with real-life driving behavior, where most daily trips tend to be relatively short, such as commuting or local travel.

The delay distribution is centered around zero, indicating that estimated travel times are generally accurate. Most values fall within a narrow range around zero, meaning that the difference between estimated and actual durations is typically small.

However, the distribution includes both negative and positive values:

- Negative values indicate that the estimated time was higher than the actual time (overestimation)
- Positive values indicate underestimation

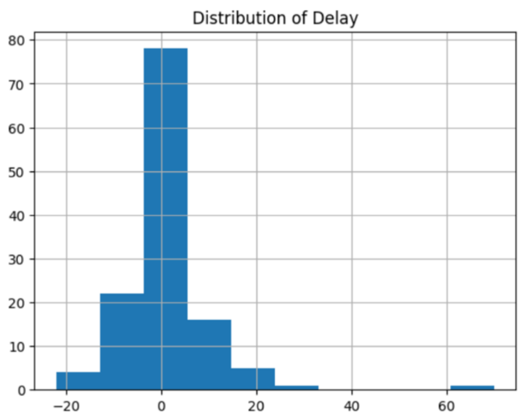

Figure 8

The presence of a few extreme values suggests that, in some cases, significant deviations occur, but these are relatively rare.

1.2.2. Correlations

A strong positive correlation (≈ 0.89) is observed between distance and actual travel duration. The scatter plot clearly shows a linear upward trend, indicating that as distance increases, travel time increases proportionally.

Most data points are tightly clustered along this trend, suggesting a consistent relationship across trips. However, a few extreme values (outliers) exist at higher distances, which may represent unusually long trips or atypical traffic conditions.

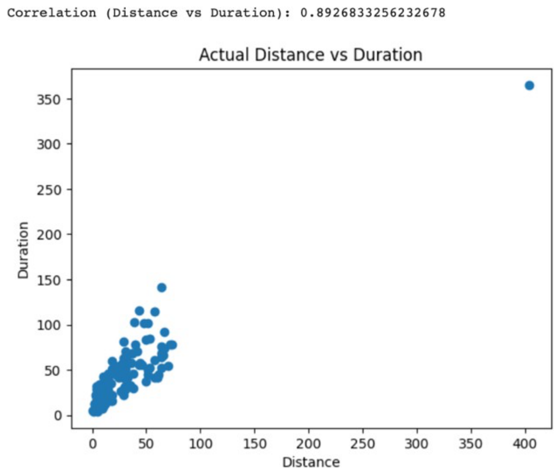

Figure 9

This result confirms that distance is the dominant factor influencing travel duration.

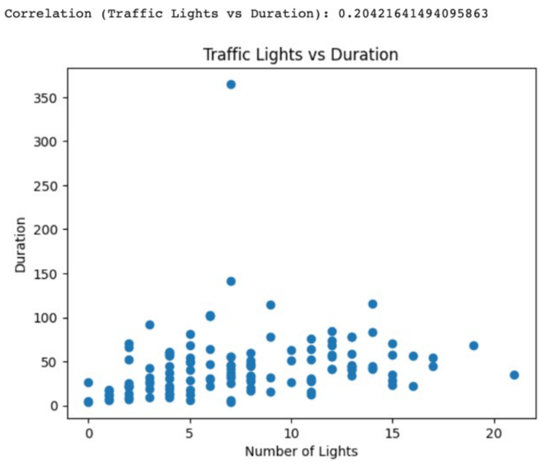

Figure 10

The correlation between the number of traffic lights and travel duration is relatively weak (≈ 0.20). The scatter plot does not show a strong linear pattern, and the data points are widely dispersed.

Although there is a slight upward tendency, the relationship is not strong enough to significantly explain variations in travel time. This suggests that traffic lights alone are not a major determinant of overall trip duration, and their effect is likely overshadowed by other variables such as distance and traffic level.

A very strong positive correlation (≈ 0.97) is observed between estimated and actual travel durations. The data points are closely aligned along a straight line, indicating a near-perfect linear relationship.

This demonstrates that estimated travel times are highly reliable predictors of actual travel duration. The consistency of this relationship suggests that navigation tools such as Google Maps are effective in capturing real-world travel conditions.

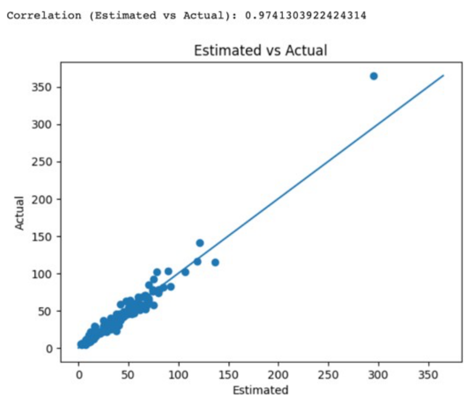

Figure 11

1.2.3. Comparison Boxplots

The boxplot shows a clear relationship between traffic level and travel duration. Trips under low traffic conditions have noticeably lower median durations and a tighter distribution, indicating more consistent and shorter travel times.

In contrast, medium and high traffic levels exhibit higher median durations and greater variability. This suggests that as traffic intensity increases, not only does travel time increase, but it also becomes less predictable.

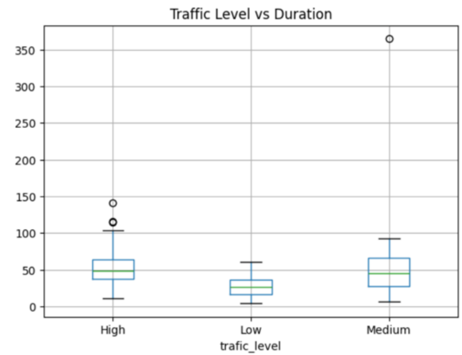

Figure 12

Additionally, the presence of outliers—especially in the medium traffic category—indicates that under certain conditions, travel duration can increase significantly even when traffic is not at its highest level. Overall, this visual pattern strongly supports the statistical finding that traffic level has a significant impact on travel duration.

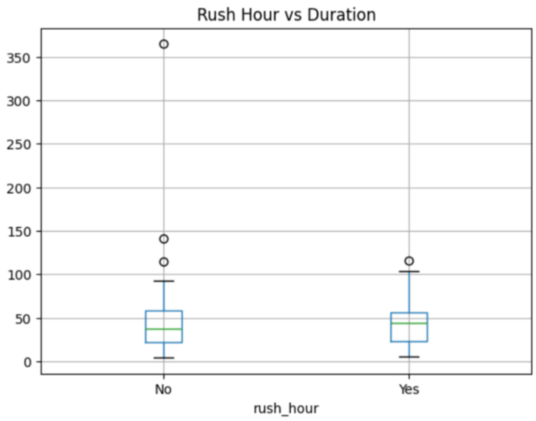

Figure 13

The boxplot comparing rush hour and non-rush hour trips does not show a clear separation between the two groups. The medians are relatively close, and the spread of the data is similar for both categories.

Although there are some extreme values (outliers), particularly in the non-rush hour group, the overall distributions overlap significantly. This indicates that rush hour alone is not a strong determinant of travel duration.

This observation is consistent with the hypothesis test results, which showed that rush hour does not have a statistically significant effect. It suggests that traffic level is a more precise and informative variable than simply categorizing trips based on rush hour.

1.2.4. Traffic Level Based Analysis

The average values provide additional quantitative evidence on how traffic conditions affect travel duration and delay.

The results shown in Figure 14 with a clear pattern.

```
Average duration by traffic level:
trafic_level
High       58.034483
Low        26.880952
Medium     51.607143
Name: actual_duration_min, dtype: float64
```

Figure 14

There is a substantial increase in travel duration as traffic level rises. Trips under medium traffic conditions take nearly twice as long as those under low traffic, while high traffic results in the longest durations.

This confirms that traffic level has a strong and direct impact on travel time, reinforcing both the visual patterns observed in boxplots and the statistical significance found in hypothesis testing.

The average delay values show a more nuanced pattern as shown in Figure 15.

```
Average delay by traffic level:
trafic_level
High       1.275862
Low       -0.285714
Medium     1.803571
Name: delay_min, dtype: float64
```

Figure 15

A negative delay under low traffic indicates that estimated travel times are slightly higher than actual durations (i.e., overestimation), meaning trips tend to be faster than expected.

In contrast, under medium and high traffic conditions, delays are positive, suggesting that actual travel times tend to exceed estimated times, although the difference remains relatively small.

2. Hypothesis Testing

Several statistical tests were conducted to validate the observed relationships:

2.1.1. Independent T-test: Rush Hour vs Non-Rush Hour

```
H0: There is no significant difference in actual travel duration between rush hour and non-rush hour trips.
H1: There is a significant difference in actual travel duration between rush hour and non-rush hour trips.
Rush hour mean: 43.785714285714285
Non-rush hour mean: 45.44705882352941
t-statistic: -0.2632160958002544
p-value: 0.792841295508658
```

Figure 16

The p-value was greater than 0.05, leading to a failure to reject the null hypothesis. This indicates that rush hour does not have a statistically significant effect on travel duration in this dataset.

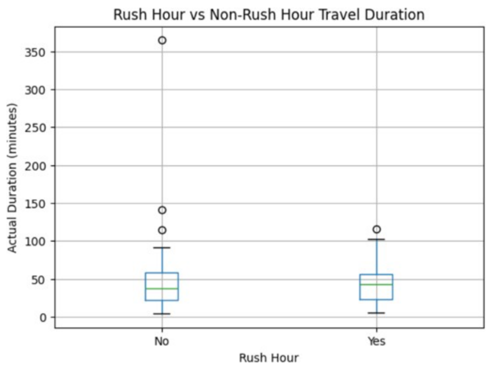

Figure 17

2.1.2. ANOVA Test: Traffic Level

```
H0: Mean actual travel duration is the same across traffic levels.
H1: At least one traffic level has a different mean actual travel duration.
F-statistic: 7.773549942802812
p-value: 0.0006598632376532952
```

Figure 18

The p-value was less than 0.05, leading to rejection of the null hypothesis. This shows that traffic level has a statistically significant impact on travel duration.

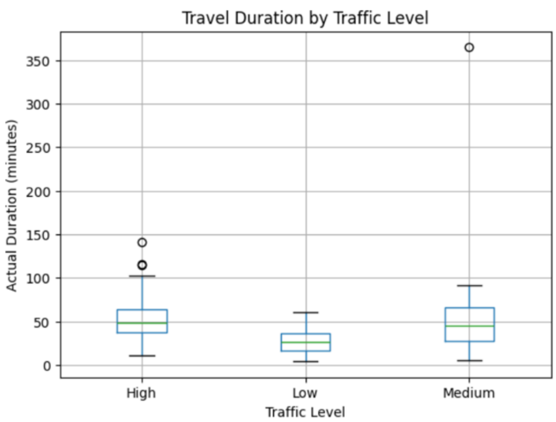

Figure 19

2.1.3. Paired T-test: Estimated vs. Actual Duration

```
H0: There is no significant difference between actual and estimated travel durations.
H1: There is a significant difference between actual and estimated travel durations.
Actual mean: 44.89763779527559
Estimated mean: 43.90551181102362
t-statistic: 1.1979862916737511
p-value: 0.23317098509478104
```

Figure 20

The test resulted in a p-value greater than 0.05, indicating no statistically significant difference between estimated and actual travel times. This suggests that Google Maps estimates are generally accurate.

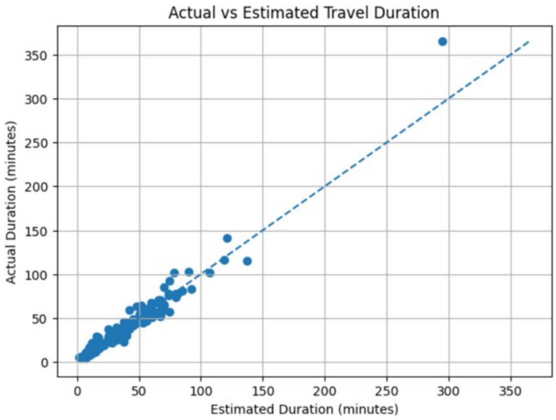

Figure 21

2.1.4. Pearson Correlation Tests: Distance and Number of Traffic Lights vs. Actual Travel Duration

Distance showed a statistically significant strong relationship with travel duration ($p < 0.05$), while the number of traffic lights also showed a statistically significant but weaker relationship.

```
H0: There is no linear relationship between distance and actual travel duration.
H1: There is a significant linear relationship between distance and actual travel duration.
Correlation coefficient: 0.8926833256232681
p-value: 4.3164259901092526e-45
```

Figure 22: Distance vs. Actual Travel Duration

```
H0: There is no linear relationship between distance and actual travel duration.
H1: There is a significant linear relationship between distance and actual travel duration.
Correlation coefficient: 0.8926833256232681
p-value: 4.3164259901092526e-45
```

Figure 23: Number of Traffic Lights vs. Actual Travel Duration

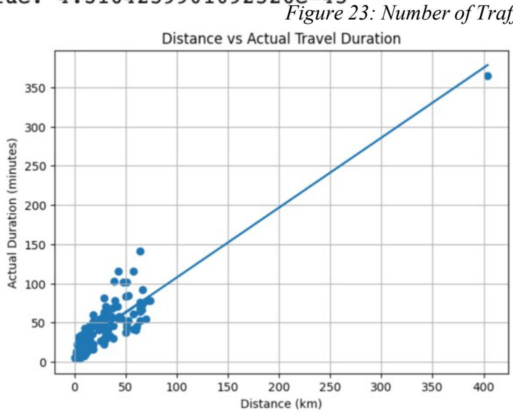

Figure 24: Distance vs. Actual Travel Duration

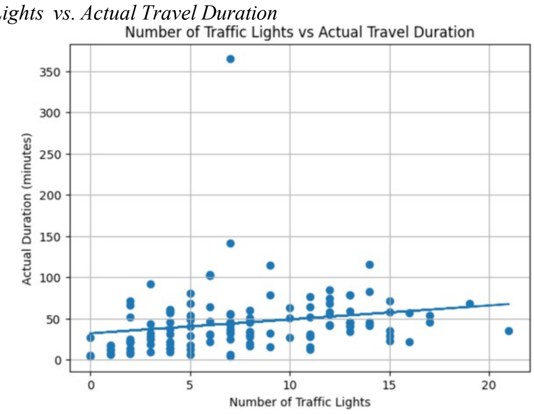

Figure 25: Number of Traffic Lights vs. Actual Travel Duration

2.1.5. One-sample T-test: Delay

```
H0: The average delay is equal to 0.
H1: The average delay is significantly different from 0.
Average delay: 0.9921259842519685
t-statistic: 1.1979862916737511
p-value: 0.23317098509478104
```

Figure 26

The average delay was not significantly different from zero, further supporting the reliability of estimated travel times.

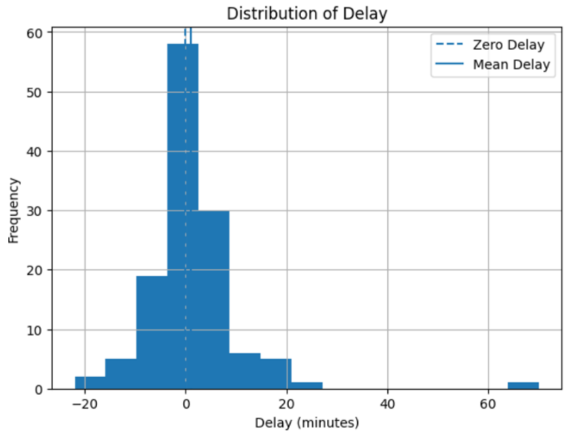

Figure 27

3. General Analysis and Interpretation

Overall, the analysis demonstrates that travel duration is primarily influenced by distance and traffic conditions. Both the exploratory data analysis and statistical tests consistently showed that distance has the strongest relationship with actual travel duration. The high Pearson correlation coefficient (approximately 0.89) indicates that longer trips naturally require longer travel times, making distance the most dominant explanatory variable in the dataset.

Traffic level was also found to be a highly important factor affecting travel duration. The ANOVA test produced a statistically significant result ($p < 0.05$), confirming that travel times differ across low, medium, and high traffic conditions. In addition, boxplots and average duration comparisons showed that travel duration increases substantially as traffic intensity rises. This suggests that traffic congestion not only increases travel time but also reduces predictability and consistency of trips.

On the other hand, the number of traffic lights showed only a weak correlation with travel duration. Although the relationship was statistically significant, the low correlation coefficient (~0.20) indicates that traffic lights alone cannot sufficiently explain variations in travel time. Their effect is likely overshadowed by stronger variables such as distance and overall traffic density.

Another important finding is related to the reliability of Google Maps estimated travel durations. The paired t-test showed no statistically significant difference between estimated and actual durations, while the correlation between them was extremely high (~0.97). These findings suggest that navigation applications are generally successful in predicting real-world travel conditions and can provide reliable duration estimates for daily transportation.

Interestingly, rush hour itself did not have a statistically significant effect on travel duration. This indicates that simply categorizing a trip as “rush hour” or “non-rush hour” may not fully capture real traffic complexity. Instead, directly measuring traffic level provides a more accurate representation of transportation conditions.

The delay analysis also supports the reliability of estimated durations. Since the average delay was not significantly different from zero, the estimation system does not systematically overestimate or underestimate travel times. However, slight positive delays observed under medium and high traffic conditions suggest that prediction errors tend to increase when traffic becomes more congested.

In general, the results indicate that travel duration prediction is highly feasible using variables such as distance, traffic level, and estimated duration. The findings obtained from EDA, hypothesis testing, and later machine learning modeling are consistent with each other, increasing the reliability and validity of the overall project conclusions.